# CA2 preparation: worked answers

Year 9 Mathematics | Chapters 3–6 | 100 marks | 120-minute worksheet

The suggested mark allocations follow the stated mark totals. Equivalent correct methods and justified alternative answers should be credited.

## Question 1 · Powers of 10 [4 marks]

**(a)** 0.063 × 1000 = **63**. [1]

**(b)** 7.2 ÷ 0.01 = **720**. [1]

**(c)** 0.0045 × 10 000 = **45**. [1]

**(d)** 540 ÷ 1000 = **0.54**. [1]

## Question 2 · Decimal calculation [3 marks]

**(a) 6** (54 ÷ 9 = 6). [1]

**(b)** 3.6 ÷ 4 = **0.9**. [1]

**(c)** 87.5 ÷ 7 = **12.5**. [1]

## Question 3 · Percentages [3 marks]

**(a)** 0.15 × 360 = **54**. [1]

**(b)** RM240 × 1.12 = **RM268.80**. [1]

**(c)** RM450 × 0.92 = **RM414**. [1]

## Question 4 · Data collection and sources [3 marks]

**(a) Continuous**, because time can take any value within a range. [1]

**(b) Primary data**; the teacher collects it directly for this investigation. [1]

**(c) Secondary data**; it was collected previously by someone else. [1]

## Question 5 · Linear equations [3 marks]

**(a)** $x - 3 = 7$, so $\mathbf{x = 10}$. [1]

**(b)** $7x - 3x = 33 - 5$; $4x = 28$; $\mathbf{x = 7}$. [2]

Marking: M1 for collecting terms; A1 for $x = 7$.

## Question 6 · Linear inequalities [3 marks]

**(a)** $2x \leq 16$, so $\mathbf{x \leq 8}$. [1]

**(b)** $-3x < 9$. Divide by $-3$ and reverse the sign: $\mathbf{x > -3}$. [2]

Marking: M1 for reaching $-3x < 9$; A1 for reversing the inequality.

### Question 7 · Simultaneous equations [3 marks]

Add the equations: 2x = 16, hence x = 8. Substitute: y = 12 − 8 = 4. **x = 8, y = 4**. [3]

Marking: M1 elimination; A1 x = 8; A1 y = 4.

### Question 8 · Basic angle facts [2 marks]

**(a)** x = 180° − 137° = **43°**. [1]

**(b)** y = 180° − 58° − 67° = **55°**. [1]

### Question 9 · Regular polygons [2 marks]

**(a)** (6 − 2) × 180° ÷ 6 = **120°**. [1]

**(b)** 360° ÷ 45° = **8 sides**. [1]

### Question 10 · Pythagoras' theorem [2 marks]

$h^2 = 8^2 + 15^2 = 64 + 225 = 289$; h = √289 = **17 cm**. [2]

Marking: M1 Pythagoras; A1 length.

### Question 11 · Sampling and bias [2 marks]

**(a)** Pupils waiting for the bus are more likely to travel by bus, so other travel methods are underrepresented. [1]

**(b)** Choose a random sample from the **whole Year 9** student list (or an appropriately stratified random sample). [1]

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### Question 12 · Compound percentage change [4 marks]

**(a)** 320 × 1.15 × 0.90 = **RM331.20**. [2]

Marking: M1 correct multipliers; A1 final price.

**(b)** Change = 331.20 − 320 = RM11.20; (11.20 ÷ 320) × 100 = **3.5% increase**. [2]

Marking: M1 percentage change; A1 3.5% increase.

### Question 13 · Bounds for perimeter and area [4 marks]

**(a)** Lower bounds: length = 8.55, width = 4.25. Minimum perimeter = 2(8.55 + 4.25) = **25.6 cm**. [2]

Marking: M1 correct lower bounds; A1 perimeter.

**(b)** Upper bounds: length = 8.65, width = 4.35. Area < 8.65 × 4.35 = **37.6275 cm²**. [2]

Marking: M1 correct upper bounds; A1 area bound.

### Question 14 · Forming equations [4 marks]

**(a)** Adult ticket = c + 10. The equation is **3(c + 10) + 2c = 70**. [2]

Marking: B1 adult ticket expression; B1 correct equation.

**(b)** 3c + 30 + 2c = 70; 5c = 40; c = 8. Adult = 8 + 10 = 18. **Child RM8; adult RM18**. [2]

Marking: M1 solve for c; A1 both costs.

### Question 15 · Simultaneous equations [4 marks]

From 2x − y = 6, y = 2x − 6. Substitute: 3x + 2(2x − 6) = 23; 7x = 35, hence x = 5. Then y = 2(5) − 6 = 4. **x = 5, y = 4**. [4]

Marking: M1 valid elimination or substitution; M1 equation in one unknown; A1 x = 5; A1 y = 4.

### Question 16 · Compound inequalities [3 marks]

**(a)** Subtract 4: $-6 \leq 3x < 9$. Divide by 3: $\mathbf{-2 \leq x < 3}$. [2]

Marking: M1 subtract 4 throughout; A1 final range.

**(b) −2, −1, 0, 1, 2**. [1]

### Question 17 · Exterior and interior angles [3 marks]

**(a)** 360 ÷ 18 = **20 sides**. [1]

**(b)** 180° − 18° = **162°**. [1]

**(c)** There are 20 equal exterior angles, and 20 × 18° = **360°**; alternatively, a full turn around the polygon is 360°. [1]

### Question 18 · Pythagoras in context [3 marks]

The wall, ground and ladder form a right-angled triangle. $d^2 = 6.5^2 - 6.0^2 = 42.25 - 36 = 6.25$. d = **2.5 m**. [3]

Marking: B1 right-angle/Pythagoras reasoning; M1 subtraction of squares; A1 2.5 m.

### Question 19 · Ruler-and-compass construction [2 marks]

Construct a perpendicular at O to ray OA to make 90°. Bisect the right angle with compass arcs to obtain the **45° ray**. Leave the construction arcs visible. [2]

Marking: B1 correct perpendicular construction; B1 correct bisection and ray.

### Question 20 · Survey sampling and leading questions [3 marks]

**(a)** Only library users at lunchtime are sampled, so the sample may not represent all Year 9 pupils. [1]

**(b)** The wording **“Don’t you agree”** is leading and encourages agreement. [1]

**(c)** For example: **“Would you prefer the school day to begin earlier, at the same time, or later?”** [1]

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### Question 21 · Bounds and checking a claim [5 marks]

**(a)** Lower bounds: 12.75 cm and 7.35 cm. Minimum possible area = 12.75 × 7.35 = **93.7125 cm²**. As 93.7125 < 94.0, the claim is **not guaranteed**. [3]

Marking: B1 lower bounds; M1 area calculation; A1 conclusion.

**(b)** Upper bounds: 12.85 cm and 7.45 cm. Perimeter < 2(12.85 + 7.45) = **40.6 cm**. [2]

Marking: M1 correct upper bounds; A1 40.6 cm.

## Question 22 · Reverse compound percentages [5 marks]

**(a)** Let the original price be P. P × 0.80 × 1.05 = 504; 0.84P = 504; P = 504 ÷ 0.84 = **RM600**. [3]

Marking: M1 multiplier 0.80; M1 multiplier 1.05 and equation; A1 RM600.

**(b)** The net multiplier is 0.80 × 1.05 = 0.84, so the final price is 84% of the original: **16% decrease**. [2]

Marking: M1 net multiplier; A1 percentage and direction.

## Question 23 · Simultaneous equations in context [5 marks]

**(a)** Let a be an adult ticket price and c a child ticket price. **3a + 4c = 102** and **5a + 2c = 114**. [2]

Marking: B1 each correct equation.

**(b)** Double the second equation: 10a + 4c = 228. Subtract first: 7a = 126 so a = 18. Substitute into 5a + 2c = 114: 90 + 2c = 114, c = 12. **Adult RM18; child RM12**. [3]

Marking: M1 valid elimination; A1 a = 18; A1 c = 12.

## Question 24 · Inequalities with consecutive integers [5 marks]

**(a)** The integers are x − 1, x and x + 1; so **45 < (x − 1) + x + (x + 1) ≤ 72**, or 45 < 3x ≤ 72. [2]

Marking: B1 terms; B1 correct compound inequality.

**(b)** 45 < 3x ≤ 72 ⇒ **15 < x ≤ 24**. [2]

Marking: M1 divide all parts by 3; A1 solution.

**(c)** **16, 17, 18, 19, 20, 21, 22, 23, 24**. [1]

## Question 25 · Interpreting survey evidence [5 marks]

**(a)** (40 ÷ 50) × 100 = **80%** of respondents. [1]

**(b)** **Selection bias:** pupils attending voluntary revision may have different views from other Year 9 pupils. **Non-response bias:** 30 of the 80 pupils surveyed did not reply, so their views are unknown. [2]

Marking: B1 each valid, distinct reason.

**(c)** Select pupils **randomly from the entire Year 9 cohort** (or stratify randomly across sets). Ask a **neutral question**, e.g. “How useful do you find Mathematics homework?” with balanced response choices. [2]

Marking: B1 appropriate sample; B1 neutral question.

### Question 26 · Pythagoras combined with percentages [5 marks]

**(a)** Height difference = 14 − 5 = 9 m. The triangle has sides 12 m and 9 m. d = √(12² + 9²) = √225 = **15 m**. [3]

Marking: B1 height difference; M1 Pythagoras; A1 15 m.

**(b)** Length = 15 × 1.08 = 16.2 m; cost = 16.2 × 18 = **RM291.60**. [2]

Marking: M1 8% extension; A1 cost.

### Question 27 · Proportionate sampling [5 marks]

**(a)** Total = 96 + 64 + 40 = 200. Blue: (96/200) × 25 = **12**. Green: (64/200) × 25 = **8**. Red: (40/200) × 25 = **5**. [3]

Marking: B1 each correct allocation.

**(b)** Assign each pupil in a house a number and use a random number generator to choose the required number without replacement. [1]

**(c)** It would introduce **selection bias** because high-attaining pupils may not represent all pupils in each house. [1]

### Question 28 · Construction of 30° [5 marks]

**(a)** With the compass radius fixed, draw an arc centred on O cutting OA at P. Draw an arc from P with the same radius to meet the first arc at B. Draw OB to form **60°**. [2]

Marking: B1 correct equilateral-arc construction; B1 correct ray.

**(b)** From points on OA and OB at equal distances from O, draw equal-radius arcs meeting inside the angle. Join O to their intersection, giving the **30° bisector**. [2]

Marking: B1 valid bisection arcs; B1 correct ray.

**(c)** Because OC bisects the 60° angle, ∠BOC = 60° ÷ 2 = **30°**. [1]

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### Coverage and weighting check

Chapter 3: Decimals, percentages and rounding — **28 marks**

Chapter 4: Equations and inequalities — **30 marks**

Chapter 5: Angles and constructions — **24 marks**

Chapter 6: Statistical investigations — **18 marks**

Difficulty distribution: **30 foundational + 30 application + 40 challenge = 100 marks**.